

Chapter 20

Electromagnetic waves

20.1 Lorentz gauge and the wave equation

Start from the Maxwell equations with source

$$\partial_\mu F^{\mu\nu} = \frac{4\pi}{c} j^\nu \quad (20.1)$$

Writing $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$, this becomes

$$\square A^\nu - \partial^\nu (\partial_\mu A^\mu) = \frac{4\pi}{c} j^\nu \quad (20.2)$$

We kill the second term by choosing the *Lorentz gauge*

$$\partial_\mu A^\mu = 0 \quad (20.3)$$

The gauge condition reads

$$\partial_0 A^0 = -\vec{\nabla} \cdot \vec{A} \Rightarrow A^0(t, \vec{x}) = A^0(0, \vec{x}) - \int_0^t \vec{\nabla} \cdot \vec{A}(t', \vec{x}) dt' \quad (20.4)$$

determining A^0 in terms of $\vec{A}$ – eliminating one of the four degrees of freedom. We are left with

$$\square A^\nu = \frac{4\pi}{c} j^\nu \quad (20.5)$$

the wave equation with a source.

20.2 Residual gauge freedom

The Lorentz gauge does not fix the gauge completely: the residual transformations

$$A'_\nu = A_\nu + \partial_\nu \Lambda \quad \text{with} \quad \square \Lambda = 0 \quad (20.6)$$

preserve (20.3). This freedom can be used to set a component to zero, for example

$$A^3(0, \vec{x}) = 0 \quad , \quad \partial_0 A^3(0, \vec{x}) = 0 \quad (20.7)$$

as follows: from

$$A^{3'} = A^3 + \partial^3 \Lambda \quad (20.8)$$

we need

$$\begin{aligned} A^{3'} &= 0 = A^3(0, \vec{x}) - \partial_3 \Lambda(0, \vec{x}) \\ \partial_0 A^{3'} &= 0 = \partial_0 A^3(0, \vec{x}) - \partial_3 \partial_0 \Lambda(0, \vec{x}) \end{aligned} \quad (20.9)$$

and we adjust Λ and $\partial_0 \Lambda$ as needed. Furthermore, the $\nu = 0$ equation is a constraint

$$-\vec{\nabla}^2 A^0 - \partial_i \partial^0 A^i = \frac{4\pi}{c} j^0 \quad (20.10)$$

(it involves only a single time derivative). Altogether, we are left with 2 degrees of freedom in A^μ .

20.3 Waves in empty space

In empty space,

$$\partial_\mu F^{\mu\nu} = 0 \Rightarrow \square A^\nu = 0 \quad \text{and} \quad \partial_\mu A^\mu = 0 \quad (20.11)$$

Write the field as a Fourier transform

$$A^\nu(x) = \frac{1}{(2\pi)^2} \int e^{i k x} \tilde{A}^\nu(k) d^4 k \quad (20.12)$$

so that the two conditions become

$$k^2 \tilde{A}^\nu(k) = 0 \quad , \quad k_\mu \tilde{A}^\mu(k) = 0 \quad (20.13)$$

The first implies

$$\tilde{A}^\nu(k) = \delta(k^2) \varepsilon^\nu(k) = \frac{1}{2\omega} (\delta(k^0 - \omega) \varepsilon^\nu(k) + \delta(k^0 + \omega) \varepsilon^{\nu*}(k)) \quad (20.14)$$

which gives

$$A^\nu = \frac{1}{(2\pi)^2} \int \frac{d^3k}{2\omega} (e^{ikx} \varepsilon^\nu(k) + \text{c.c.}) \quad (20.15)$$

with

$$k^0 \equiv \omega = |\vec{k}| \quad , \quad kx \equiv k_\mu x^\mu \quad (20.16)$$

This is a superposition of elementary waves with amplitude $\varepsilon^\nu(k)$ satisfying

$$k_\nu \varepsilon^\nu = 0 \quad (20.17)$$

because $\partial_\mu A^\mu = 0$. In addition, gauge invariance acts on the amplitude as

$$\varepsilon^\nu \rightarrow \varepsilon^\nu + \lambda k^\nu \quad (20.18)$$

Conditions (20.17) and (20.18) together leave 2 polarizations (instead of 4). The field strength is then

$$F^{\mu\nu}(t, \vec{x})[?] = \frac{i}{2(2\pi)^2} \int \frac{d^3k}{\omega} (e^{ikx} (k^\mu \varepsilon^\nu - k^\nu \varepsilon^\mu) + \text{c.c.}) \quad (20.19)$$

20.4 Elementary waves

An elementary wave is

$$A^\mu(x) = \varepsilon^\mu e^{ikx} + \text{c.c.} \quad (20.20)$$

Computing the corresponding fields, one finds

$$\hat{k} \cdot \vec{E} = \hat{k} \cdot \vec{B} = 0 \quad , \quad \vec{E} \cdot \vec{B} = 0 \quad , \quad E = B \quad , \quad \vec{B} = \hat{k} \times \vec{E} \quad (20.21)$$

For example, take $k^\mu = (\omega, 0, 0, \omega)$ and decompose the amplitude into longitudinal and transverse pieces

$$\varepsilon_L^\mu = (\varepsilon^0, 0, 0, \varepsilon^0) \quad , \quad \varepsilon_T^\mu = (0, \varepsilon^1, \varepsilon^2, 0) \quad (20.22)$$

Under the shift (20.18),

$$\begin{aligned} \varepsilon^\mu \rightarrow \varepsilon^\mu + \lambda k^\mu &= (\varepsilon^0 + \lambda\omega, \varepsilon^1, \varepsilon^2, \varepsilon^0 + \lambda\omega) \\ &= \varepsilon_T^\mu + \left(1 + \frac{\lambda\omega}{\varepsilon^0}\right) \varepsilon_L^\mu \end{aligned} \quad (20.23)$$

and we can choose λ so that the coefficient of ε_L^μ vanishes: we are left with ε_T^μ – transverse waves.

20.5 Polarization

Write the electric field of an elementary wave as

$$\vec{E} = \frac{\vec{\mathcal{E}}}{2} e^{i k x} + \frac{\vec{\mathcal{E}}^*}{2} e^{-i k x} \quad , \quad \hat{k} \cdot \vec{\mathcal{E}} = 0 \quad (20.24)$$

and decompose the complex amplitude as

$$\vec{\mathcal{E}} = \vec{V}_1 + i \vec{V}_2 \Rightarrow \vec{V}_1 \cdot \hat{k} = \vec{V}_2 \cdot \hat{k} = 0 \quad (20.25)$$

so that $\vec{V}_1$ and $\vec{V}_2$ span the transverse two-dimensional space. If $\vec{V}_1 \parallel \vec{V}_2$, we have *linear polarization*

$$\vec{\mathcal{E}} \propto \vec{V}_{1,2} \quad (20.26)$$

If $\vec{V}_1 \perp \vec{V}_2$, we have *circular polarization*

$$\hat{k} \times \vec{\mathcal{E}} = \pm i \vec{\mathcal{E}} \quad , \quad \text{e.g.} \quad \mathcal{E}^x = \pm i \mathcal{E}^y \quad (20.27)$$

– clockwise if $\vec{V}_2 \times \vec{V}_1 \parallel \hat{k}$, counterclockwise if anti-parallel. Otherwise, we have elliptic polarization.

The energy-momentum tensor of the elementary wave is

$$T_{EM}^{\mu\nu} = \frac{k^\mu k^\nu}{8\pi \omega^2} (E^2 + B^2) c = \frac{k^\mu k^\nu c}{4\pi \omega^2} E^2 \quad (20.28)$$

with Poynting vector

$$S^i = \frac{k^i c}{4\pi \omega} E^2 \quad (20.29)$$

These relations for elementary waves hold for general waves *far away* from sources.

Note: the two polarization states correspond to spin ± 1 in quantum mechanics, while the longitudinal, unphysical polarization would correspond to spin 0 – the photon is a spin 1 particle in quantum mechanics.